

MD. MEHEDI HASAN

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RESEARCH INTERESTS

Behavioral machine learning, social engineering detection, interpretable AI, socio-economic modeling, continual learning, and geospatial risk prediction. Focused on building AI systems that are explainable, fair, and grounded in real-world human behavior.

EDUCATION

Dhaka International University

Sep 2023 – Dec 2027

Bachelor of Science in Computer Science and Engineering

CGPA: 3.76 / 4.00

BAF Shaheen College Jashore

Sep 2020 – Dec 2022

Higher Secondary Certificate

GPA: 4.75 / 5.00

PROJECTS

SocialAI – Autonomous Engagement Framework

- Engineered a distributed, scalable platform integrating LLM agents and resilient proxy routing for automated social media content generation and task execution at scale.

SafeRoads – Geospatial Risk Prediction

- Built a system identifying high-risk road segments using traffic, weather, and accident data.
- Supports urban planning and risk-aware navigation.
- Methods: Geospatial ML, XGBoost, Spatial Clustering*

MeteorShield – NASA Space Apps Challenge 2025

- Developed a planetary defense visualization platform using NASA NEO datasets.
- Champion (Barisal Division), Global Nominee, Honorable Mention**
- Focused on data storytelling and public awareness.

SaveFood – AI for Food Waste Reduction

- Trained time-series ML models to predict food spoilage from IoT sensor data (F1-score: 0.89).

AgroHub – IoT & ML Smart Agriculture

- Real-time farm monitoring and ML-driven decision support system for smallholder farmers to optimize yield.

DevBurn — Developer Analytics Engine

- A robust TypeScript-based full-stack engine designed to analyze developer workflows, integrating GitHub services with burnout tracking algorithms.

AI Learning Platform

- An AI-powered supplementary educational platform built to enhance university coursework with intelligent assistance and interactive tracking.

CryptoTracker

- A real-time cryptocurrency portfolio management application for tracking live market data and analyzing asset performance.

PUBLICATIONS

Accepted (11)

1. **Hasan, M. M., Rakib, R., Molla, M. A., Borhan, R., Based, M. A. A Socio-Economic Machine Learning Framework for Predicting Programmer Retention.** *BIM 2025, Taylor & Francis.* [\[Accepted\]](#)
2. **Hasan, M. M., Mahin, A. A., Chakraborty, S., Afrose, M., Mia, M. A., Based, M. A. Fraud Detection with DistilBERT: A Transformer-Based Approach to Behavioral Banking Sequences.** *BIM 2025, Taylor & Francis.* [\[Accepted\]](#)
3. **Hasan, M. M., Mahin, A. A., Afrose, M., Based, M. A. MEWC-RL++: A Modular and Adaptive Continual Reinforcement Learning Framework for Automation Systems.** *ECCT 2026, Taylor & Francis.* [\[Accepted\]](#)
4. **Mrida, M. A. R., Hasan, M. M., Ausaf, S. M. N., Ray, P. C., Based, M. A. Impact of Social Media Usage on Academic Performance: A Hybrid Machine Learning and SEM Approach.** *ECCT 2026, Taylor & Francis.* [\[Accepted\]](#)
5. **Biswas, S., Hasan, M. M., Rakib, R., Mahin, A. A., Afrose, M., Based, M. A. Lightweight Deep Learning for Urban Traffic Intelligence: A MobileNetV2-YOLOv8 Pipeline with Adaptive Resolution Processing.** *ECCT 2026, ACSR, Springer Nature.* [\[Accepted\]](#)
6. **Asif, M., Hasan, M. M., Based, M. A. Federated and Explainable Machine Learning Framework for Cyber Attack Detection in Smart Grids.** *ECCT 2026, Taylor & Francis.* [\[Accepted\]](#)
7. **Hasan, M. M., Hossain, M. A., Mahin, A. A., Khatun, F. Student-YOLOv8: A YOLO-Based Deep Learning Approach for Real-Time Student Detection and Face Recognition in Classroom Attendance Monitoring Systems.** *IUB Journal of Science and Engineering (AJSE).* [\[Accepted\]](#)
8. **Rion, A. M., Pallob, M. M. I., Rakib, R., Molla, M. A., Hasan, M. M. Cross-Cultural Early Detection of Suicidal Ideation in Adolescents: An Ensemble Machine Learning Approach Using GSHS Data.** *IEEE TENCON 2026.* [\[Accepted\]](#)
9. **Molla, M. A., Rakib, R., Hasan, M. M., Pallob, M. I., Rion, A. M., Based, M. A. A Hybrid Explainable AI Framework for Concurrent Earthquake Magnitude Estimation and Severity Classification.** *MIET 2026, Springer Nature LNNS.* [\[Accepted\]](#)
10. **Molla, M. A., Rakib, R., Pallob, M. I., Hasan, M. M., Rion, A. M. FairCF: Fair Counterfactual Explanations for Student Academic Performance Prediction.** *ICAIMS 2026, IEEE.* [\[Accepted\]](#)
11. **Ray, D. D., Hasan, M. M., Ausaf, S. M. N., Based, M. A., Rahman, M. M. Toward an AI-Assisted Dyslexia Detection Framework for Bangla-Speaking Children: A Cross-**

Under Review (14)

12. **Hasan, M. M., Pallob, M. M. I., Based, M. A. Digital Transformation for Sustainable Universities: Modeling the Convergence of MIS, IoT, and Governance.** [Under Review – Journal]
13. **Hasan, M. M., Mahin, A. A., Afrose, M., Based, M. A. War, Media Bias, and Socio-Economic Consequences: A Centennial Analysis of the Narrative-Reality Index (1914–2026).** [Under Review – Journal]
14. **Hasan, M. M., Rahman, M. M., Mrida, M. A. R., Based, M. A. A Hybrid Explainable Machine Learning Framework for Customer Lifetime Value Prediction and Its Socio-Economic Implications.** [Under Review]
15. **Rakib, R., Hasan, M. M., Molla, M. A., Based, M. A. A Relational Database-Driven Real-Time Cryptocurrency Portfolio Management System with Automated ETL Integration.** [Under Review]
16. **Rahman, M. M., Hasan, M. M., Based, M. A. SafeRoads: AI-Driven Proactive Risk Synthesis and Data-Centric Infrastructure Audits for Road Safety.** [Under Review]
17. **Molla, M. A., Hasan, M. M., Rakib, R., Pallob, M. I., Borhan, R., Based, M. A. Bridging the Urban-Rural Digital Divide in Bangladeshi CS Education: Counterfactual XGBoost Explanations, Geography-Moderated SEM and Infrastructure Policy Simulations.** [Under Review]
18. **Hasan, M. M., Molla, M. A., Rakib, R., Pallob, M. I., Borhan, R., Based, M. A. Early Identification of High Risk CS Dropout Students in Bangladesh: A Pseudo Survival XG-Boost Framework with SHAP Interpretability, Structural Equation Modeling and Policy Simulations.** [Under Review]
19. **Rakib, R., Hasan, M. M., Molla, M. A., Pallob, M. I., Borhan, R., Based, M. A. Quantifying Intrinsic Motivation in Computer Science Career Intent: A Multi-Method XAI Framework, Full-Mediation SEM and Incentive Policy Simulations.** [Under Review]
20. **Afrose, M., Hasan, M. M., Mahin, A. A., Based, M. A. Fairness-Aware HIV Risk Prediction in Bangladesh Using Behavioral and Explainable AI.** [Under Review]
21. **Mahin, A. A., Hasan, M. M., Rakib, R., Molla, M. A., Based, M. A. A Modular Framework for Continual Reinforcement Learning in Dynamic Robotic Environments.** [Under Review]
22. **Mahin, A. A., Hasan, M. M., Afrose, M., Mia, M. A., Based, M. A. Interpretable Machine Learning for Depression Risk Classification Among Bangladeshi University Students.** [Under Review]
23. **Molla, M. A., Hasan, M. M., Rakib, R., Pallob, M. M. I., Borhan, R., Based, M. A. Gender Based Predictive Persistence in Computer Science Education: A Disaggregated SHAP Analysis, Gender Moderated SEM and Policy Interventions in Bangladesh.** [Under Review]

24. **Hasan, M. M.**, Molla, M. A., Rakib, R., Pallob, M. M. I., Borhan, R., Based, M. A. **Uncovering Socioeconomic Archetypes in Bangladeshi CS Undergraduates: K-Modes Clustering, Latent Class SEM and Targeted Archetype Policy Simulations.** [Under Review]
25. Molla, M. A., **Hasan, M. M.**, Rakib, R., Borhan, R., Based, M. A. **Urban–Rural Inequality in Tech Career Intention: A Machine Learning Perspective.** [Under Review]

Ongoing Research (4)

26. **Hasan, M. M.**, Rakib, R., Molla, M. A., Based, M. A. **Women’s Psychology and Its Socioeconomic Impact on Male Partners and Relationships.** [Ongoing]
27. **Hasan, M. M.**, Molla, M. A., Rakib, R., Based, M. A. **THE DUAL PARADOX: AI Investment and Workforce Displacement in Global Corporations.** [Ongoing]
28. Afrose, M., **Hasan, M. M.**, Based, M. A. **MEWC-LLM: Memory-Efficient Weight Consolidation for Large Language Models – Adaptive Stability–Plasticity Optimization for Factual Continual Learning.** [Ongoing]
29. Samira, T., **Hasan, M. M.**, Based, M. A. **The Use of AI and Its Negative Impacts on Students and Socioeconomic Development.** [Ongoing]

ACHIEVEMENTS

NASA International Space Apps Challenge 2025 — Champion (Barisal), Global Nominee, Honorable Mention — [Team Polaris](#)
National Data Analytics Competition (NDAC) 2025 — Participant
Eastern Bank PLC Technovation’25 National Hackathon 2025 — Participant
BUET CSE FEST National Hackathon 2026 – Participant

PROFESSIONAL AFFILIATIONS

IEEE — Student Member (Valid through Dec 2026)
IEEE Computer Society — Student Member (Valid through Dec 2026)
BASIS – Student Member

LEADERSHIP & COMMUNITY

DIU Computer Programming Club - President	<i>April 2026 – Present</i>
DIU CSE Speakers Club — General Secretary	<i>May 2025 - March 2026</i>
DIU Career Development Club — Head, Research and Training Wing	<i>Aug 2025 – Dec 2025</i>
BASIS Students’ Forum (DIU Chapter) — Executive Member	<i>Sep 2024–Oct 2025</i>

TECHNICAL SKILLS

Languages: Python, C, Java, SQL
ML Frameworks: Scikit-learn, TensorFlow, PyTorch, Auto-sklearn, LLMs / GenAI
Automation & Bots: Workflow automation, intelligent bots, web automation (Selenium, Playwright), API automation, task scheduling, RPA concepts, proxy engineering, logging & exception handling
Data Tools: Pandas, NumPy, Matplotlib, Seaborn
Dev Tools: Git, GitHub, VS Code, Google Colab, Kaggle
Specialized: Interpretable AI (SHAP), geospatial analysis, time-series forecasting, AutoML

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